

UTME 2016 BIOLOGY QUESTIONS

1. Which of the following structures is a protective adaptive feature of the Agama Lizard to the environment?
 - A. Nuchal crest
 - B. Claws
 - C. Scaly skin
 - D. Gular fold.
2. Which of the following adapts an insect for feeding?
 - A. suitable mouthparts
 - B. paired antennae
 - C. segmented body
 - D. jointed appendages
3. Which of the following results from the cross between Yy and Yy?
 - A. 2Yy-2yy
 - B. 2Yy:yy:YY
 - C. YY:2Yy:yy
 - D. YY: Yy:2yy
4. Which of the following is NOT part of the carbon cycle?
 - A. Organic carbon
 - B. Decomposition
 - C. Nitrates formation
 - D. Photosynthesis
5. I. Tissues
 II. System
 III. Cell
 IV. Organs
 Which of the above is the level of organization of a leaf?
 - A. IV
 - B. I.
 - C. III.
 - D. II.
6. In cellular respiration, energy is stored in the form of
 - A. heat energy
 - B. adenosine diphosphate
 - C. adenosine monophosphate
 - D. adenosine triphosphate
7. The principal organ for the manufacture of food in autotrophy is the
 - A. root hair
 - B. growing root
 - C. mature fruit
 - D. green leaf
8. A disease that results from lack of iodine in the diet of humans is
 - A. beriberi
 - B. scurvy
 - C. rickets
 - D. goiter
9. The process whereby some organism with certain favourable features get established in an area is
 - A. gene mutation
 - B. dispersal
 - C. overcrowding
 - D. natural selection
10. The rise and fall of ocean water during the day is referred to as
 - A. gravity
 - B. a pull
 - C. tide
 - D. zone
11. Which of the following is a producer in an aquatic habitat?
 - A. Nymphaea
 - B. Dryopteris
 - C. planarian
 - D. Similium
12. The relationship that exist between a shark and Remora is
 - A. parasitism
 - B. commensalism
 - C. saprophytism
 - D. symbiosis

13. I. Tissue
II. System
III. Cell
IV Organ
The correct sequence of increasing level of complexity is
A. IV-II-III
B. I-II-III-IV
C. IV-III-I-II
D. III-I-IV-II

14. Which of the following is not an inheritable disease?
A. Poliomyelitis
B. Sickle-cell anaemia
C. Mental illness
D. Haemophilia

15. Which of the finger print types occur most frequently in the population of human beings
A. Double-loop
B. Whorl
C. Arch
D. Loop

16. Beriberi results from a deficiency of
A. vitamin A
B. vitamin E
C. vitamin B
D. vitamin C

17. Bacteria which add atmospheric nitrogen to the soil are
A. putrefying bacteria
B. nitrifying bacteria
C. nitrogen fixing bacteria
D. denitrifying bacteria

18. The spines of the hedgehog is an adaptive features for
A. Courtship
B. defence
C. water conservation
D. obtaining food

19. The dental formula of carnivores is represented by

A. $\frac{I\ 0}{3}, \frac{C\ 1}{1}, \frac{pm\ 4}{4}, \frac{m\ 2}{3}$
B. $\frac{I\ 0}{2}, \frac{C\ 1}{1}, \frac{pm\ 4}{4}, \frac{m\ 2}{4}$
C. $\frac{I\ 2}{3}, \frac{C\ 2}{1}, \frac{pm\ 3}{4}, \frac{m\ 2}{3}$
D. $\frac{I\ 3}{3}, \frac{C\ 1}{1}, \frac{pm\ 4}{4}, \frac{m\ 2}{2}$

20. Which of the following instruments is used to measure temperature?
A. Thermometer
B. Hygrometer
C. Anemometer
D. Hydrometer

21. In human, puffiness and water retention in the body is a possible symptom of
A. bladder malfunction
B. poor digestion
C. kidney malfunction
D. obesity

22. The theory of evolution which postulates that all living organisms have a common ancestor was proposed by
A. Linnaeus
B. Darwin
C. Lamarck
D. Mendel

23. Mammals requires roughage in their food to
A. provide energy
B. slow down aging
C. ease digestion
D. prevent disease

24. Variation can occur among offspring of living organism because
A. seeds are produced by self-pollination
B. zygotes are produced by cross fertilisation
C. they are produced by binary fission
D. they are produced without fertilisation

25. The most important biotic factors which affect plants and animals in the habitat are
- temperature and rainfall
 - temperature and turbidity
 - salinity and relative humidity
 - rainfall and relative humidity
26. The lowest unit of classification is the
- Kingdom
 - class
 - phylum
 - species
27. Two important processes involved in the absorption and transport of materials in plants are
- flaccidity and turgidity
 - diffusion and plasmolysis
 - plasmolysis and capillarity
 - osmosis and diffusion
28. A series of organisms existing in an ecosystem through which energy is transformed can be referred to as
- food cycle
 - food chain
 - pyramid on numbers
 - food web
29. The cell organelle solely responsible for respiration is the
- nucleus
 - nucleolus
 - endoplasmic reticulum
 - mitochondrion
30. In which part of Nigeria are Mangrove swamps found?
- Chad Basin
 - Niger Delta
 - Benue Valley
 - Mambilla Plateau
31. The breeding methods that are useful in selective breeding of animals and plants are
- inbreeding and cross-breeding
 - inbreeding and hetero-breeding
 - inbreeding and out-breeding
 - inbreeding and self-breeding
32. In a small unicellular organism, diffusion is sufficient for transport because
- the surface area to volume ratio is small
 - they have lungs for diffusion
 - materials have to move over long distance
 - the surface area to volume ratio is large
33. The function of the spinal cord is to
- stand the body structure erect
 - control involuntary actions
 - transmit impulses to the brain
 - regulates developmental changes
34. The first vertebrates to venture out of water and live on land are the
- Pisces
 - Amphibians
 - Reptiles
 - Aves
35. Which of the following factors mostly determine the major biomes of the world.
- pressure and wind speed
 - temperature and wind speed
 - pressure and rainfall
 - Temperature and rainfall
36. I. Strong winds
II. high temperature
III. Dry and porous soils.
Which group of plants are specially adapted to grow under environmental conditions stated above?
- Thallophytic
 - Mesophytes
 - Xerophytes
 - Hydrophytes

37. The lowest unit of a biogeographical plant species is

- A. micro flora
- B. macro fauna
- C. micro fauna
- D. macro flora

38. Which of the following is rich source of vitamin K?

- A. Tomato
- B. Guava
- C. Milk
- D. Onion

39. Severe diarrhea, dehydration and weakness are symptoms of

- A. cholera
- B. chickenpox
- C. malaria
- D. yellow fever

40. A common characteristic found among the crustaceans is the possession of

- A. a pair of antennae
- B. a pair of walking legs on each segment
- C. four pairs of walking legs on the cephalothorax
- D. two pairs of antennae

41. In which of the following groups of invertebrates are flagella and cilia found

- A. annelids
- B. protists
- C. coelenterates
- D. Anthropods

42. Physiological variation in human population is evidence in the

- A. difference in the fingerprints
- B. physical appearance of individuals
- C. differences in height and weight
- D. ability to roll the tongue

43. In photosynthesis, oxygen is liberated during

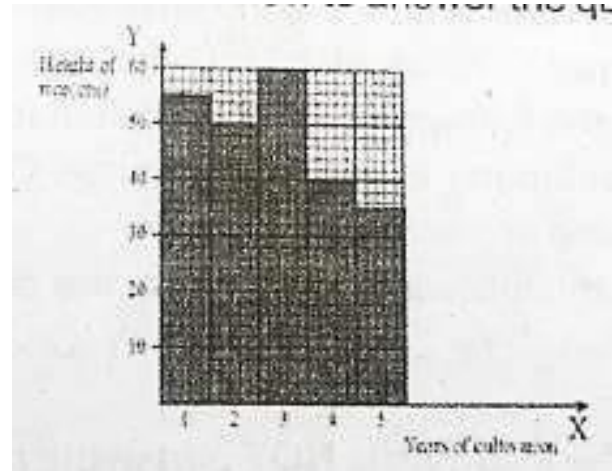
- A. conversion of energy

B. photolysis

C. splitting of carbon (IV)oxide

D. glycolysis

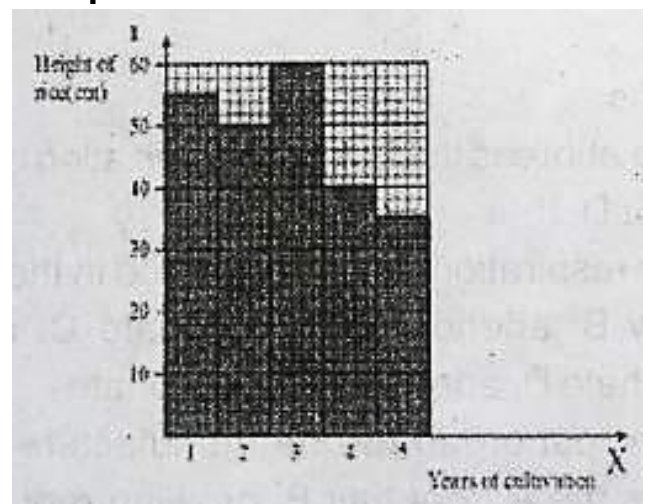
44. Use the information below to answer the question that follows



What is the total height of rice that grew within the years of cultivation?

- A. 240 cm
- B. 239 cm
- C. 340 cm
- D. 339 cm

45. Use the information below to answer the question that follows



What is the average of the heights of rice within the period of cultivation?

- A. 68cm
- B. 47.8cm
- C. 48cm
- D. 67.8cm

46. Use the diagram below to answer the question that follows

The type vertebra represented in the diagram is

- A. Atlas
- B. Lumbar
- C. Axis
- D. Sacrum

47. Use the diagram below to answer the question that follows

The structure labeled I is

- A. Centrum
- B. neural canal
- C. neural spins
- D. transverse process

ANSWER KEYS:

1. C
2. A
3. C
4. C
5. A
6. D
7. D
8. D
9. D
10. C
11. C
12. B
13. D
14. A

15. B
16. B
17. C
18. C
19. B
20. C
21. A
22. C
23. B
24. C
25. B
26. A
27. D
28. D
29. C
30. D
31. B
32. C
33. D
34. C
35. B
36. D
37. C
38. A
39. D
40. A
41. D
42. B
43. D
44. B
45. B
46. B
47. B